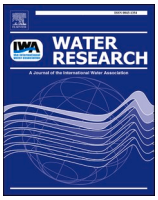
Water Research 256 (2024) 121645



Contents lists available at ScienceDirect

Water Research

journal homepage: www.elsevier.com/locate/watres





Degradation of methylmercury into Hg(0) by the oxidation of iron (II) minerals

Fuyu Xie [a,b], Qingke Yuan [a], Ying Meng [a,*], Fubo Luan [a,b,*]

[a] Key Laboratory of Drinking Water Science and Technology, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences, Beijing 100085, PR China
[b] University of Chinese Academy of Sciences, Beijing 100049, PR China

ARTICLE INFO

Keywords:
Abiotic demethylation
Light-independent
Redox fluctuations
Mercury cycling

ABSTRACT

Mercury contamination is a global concern, and the degradation and detoxification of methylmercury have gained significant attention due to its neurotoxicity and biomagnification within the food chain. However, the currently known pathways of abiotic demethylation are limited to light-induced photodegradation process and little is known about light-independent abiotic demethylation of methylmercury. In this study, we reported a novel abiotic pathway for the degradation of methylmercury through the oxidation of both mineral structural iron(II) and surface-adsorbed iron(II) in the absence of light. Our findings reveal that methylmercury can be oxidatively degraded by reactive oxygen species, specifically hydroxyl and superoxide radicals, which are generated from the oxidation of iron(II) minerals under dark conditions. Surprisingly, Hg(0) trapping experiments demonstrated that inorganic Hg(II) resulting from the oxidative degradation of methylmercury was rapidly reduced to gaseous Hg(0) by iron(II) minerals. The demethylation of methylmercury, coupled with the generation of Hg(0), suggests a potential natural attenuation process for methylmercury. Our results highlight the underappreciated roles of iron(II) minerals in the abiotic degradation of methylmercury and the release of gaseous Hg(0) into the atmosphere.

1. Introduction

Mercury is a highly toxic global contaminant that persists in the environment (Driscoll et al., 2013). Compared to inorganic mercury, methylmercury (MeHg) exhibits greater lipophilicity to cross the placenta and concentrates in the central nervous system (Trasande et al., 2005), resulting in severe health risks to humans upon ingestion of contaminated fish or rice grains (Blum, 2011; Liu et al., 2019). Consequently, there has been a growing concern involving the transformation of inorganic mercury into MeHg in the natural environment over the past decades (Hu et al., 2013; Parks et al., 2013; Tian et al., 2021). Recently, the demethylation of MeHg has gained more attention because degrading it into less toxic inorganic mercury is believed as a significant detoxification pathway. The currently known pathways of demethylation are limited to specific microbial demethylation and non-biological photo-oxidation processes. Biotic demethylation mediated by *mer* operon functions has been found in aerobic prokaryotes (Kritee et al., 2009), and *mer*-independent demethylation by certain anaerobic microorganisms or methanotrophs has been considered to be related to C1 metabolism (Barkay and Gu, 2022; Kang-Yun et al., 2022; Liu et al., 2023; Lu et al., 2017). Abiotic demethylation induced by photo-oxidation processes occurs in sunlit surface waters (Qian et al., 2014; Sellers et al., 1996; Tai et al., 2014). Reactive oxygen species (ROS) are recognized as the dominant reactive oxidants generated by sunlight, and they possess the ability to oxidize the methyl group in MeHg, resulting in the generation of inorganic mercury (Zhang and Hsu-Kim, 2010). However, abiotic degradation of MeHg in dark oxic conditions has received limited investigation, which is required for a comprehensive understanding of the biogeochemical cycling of mercury.

Iron minerals, including iron-bearing clay minerals and iron oxides, are ubiquitous in natural environments (Chen et al., 2017; Peretyazhko et al., 2012), and their oxidation and reduction cycling plays crucial roles in the transformation of various contaminants (Kappler et al., 2021; Zhao et al., 2021). In recent years, a growing body of evidence has revealed that the oxidation of both structural Fe(II) in clay minerals and adsorbed Fe(II) on iron oxides generates hydroxyl radical (˙OH) in O_2-perturbed dark subsurface environments (Xie et al., 2020; Zhang et al., 2023b). Known for its exceptional oxidizing power in nature, ˙OH has been demonstrated to oxidize both inorganic and organic

* Corresponding authors.
E-mail addresses: yingmeng@rcees.ac.cn (Y. Meng), fbluan@rcees.ac.cn (F. Luan).

https://doi.org/10.1016/j.watres.2024.121645
Received 17 February 2024; Received in revised form 25 March 2024; Accepted 18 April 2024
Available online 19 April 2024
0043-1354/© 2024 Elsevier Ltd. All rights reserved.

contaminants under dark conditions when O_2 disrupts the redox conditions (Chen et al., 2018; Cheng et al., 2020a; Sun et al., 2020; Zeng et al., 2017). However, it remains unclear whether oxidative degradation of MeHg occurs during the oxidation of Fe(II)-bearing minerals.

Additionally, Fe(II)-bearing minerals have been found to readily reduce inorganic Hg(II) to gaseous Hg(0) (Amirbahman et al., 2013; Jeong et al., 2010; Wiatrowski et al., 2009). Considering their natural reductive capacity and the generation of reactive oxidants during the oxidation process of Fe(II)-bearing minerals, the involvement and mechanism of these minerals in MeHg demethylation under dark oxic conditions are likely to be intricate. In this study, we demonstrate that the oxidative degradation of MeHg is triggered by the oxidation of both structural Fe(II) and surface-adsorbed Fe(II) under dark conditions. Subsequent reduction of inorganic Hg(II), the primary product of MeHg degradation, occurs, leading to the discharge of Hg(0). Our findings offer evidence supporting the significance of Fe(II)-bearing minerals in the cycling of Hg within fluctuating redox environments, suggesting that these fluctuations may contribute to a previously overlooked process involving the demethylation of MeHg and the discharge of gaseous Hg (0) into the atmosphere.

2. Materials and methods

2.1. Preparation of Fe(II)-bearing minerals

Mineral structural Fe(II) was prepared by reducing NAu-2, an iron-rich smectite purchased from the source clays repository of the Clay Minerals Society, with an empirical dosage of sodium dithionite in citrate-bicarbonate buffer (Luan et al., 2015). After this chemical reduction, reduced nontronite (rNAu-2) was washed once with 1 M NaCl and three times with N_2-purged ultrapure water (> 18 MΩ cm) to remove residual dithionite. A stock suspension of rNAu-2 with Fe (II)/total Fe of 38 % was stored in the anaerobic glovebox (Mikrouna) in darkness for later use. Surface-adsorbed Fe(II) was created by adding 2 mM soluble Fe(II) to 50 g/L magnetite (Aladdin) at initial pH of 7.15 (buffered by 20 mM borate) in the anaerobic glovebox. All the soluble Fe (II) was adsorbed onto Fe_3O_4 surfaces after equilibrium for overnight.

2.2. Methylmercury degradation during the oxidation of Fe(II)-bearing minerals

Methylmercury (MeHg) degradation experiments during the oxidation of 2 g/L rNAu-2 or 2 mM surface-adsorbed Fe(II) were conducted in autoclaved amber glass vials (12 mL). All vials were acid-washed and combusted at 500 °C for 0.5 h before use. About 4 mL of Fe(II)-bearing minerals suspension containing 25 nM MeHg buffered by 20 mM borate (pH 7.15 ± 0.10) was prepared in the anaerobic glovebox. The vials were sealed with polytetrafluoroethylene/silicone screw caps and kept at 150 rpm on an orbital shaker (25 ± 1 °C) under dark anoxic conditions for 3 h before oxidation, which was adequate for MeHg adsorption on mineral to reach its plateau (Fig. S1). Then, MeHg degradation experiments were started by moving the systems out from the glovebox to air. The air in the headspace (8 mL) was sufficient for the oxidation of Fe (II)-bearing minerals, and the suspension was shaken in darkness for 3 h (150 rpm, 25 ± 1 °C). Anoxic control experiments with O_2-free suspension were conducted in the anaerobic glovebox under dark conditions. To explore the involvement of hydroxyl radical ($^{\cdot}OH$) and superoxide radical ($O_2^{\cdot-}$) in MeHg degradation, quenching experiments were carried out by adding 1 M methanol and 1 mM 1,4-benzoquinone (PBQ), respectively. To examine whether Fe(II) oxidation was essential for MeHg degradation, pristine NAu-2 without chemical reduction and Fe_3O_4 alone (50 g/L), were separately mixed with 25 nM MeHg in the anaerobic glovebox. Additionally, to clarify whether oxidation of dissolved Fe(II) could induce the degradation of MeHg, especially in the assay of surface-adsorbed Fe(II), soluble Fe(II) (2 mM) without minerals was mixed with 25 nM MeHg in dark anoxic conditions. Likewise, after 3-h equilibrium, control experiments were moved out from the anaerobic glovebox and exposed to air for oxidation in the dark (150 rpm, 25 ± 1 °C).

To confirm the capability of Fe(II)-bearing minerals to reduce inorganic Hg(II) in this study, reduction experiments were carried out under dark anoxic conditions. In the reduction experiments of inorganic Hg (II), 25 nM inorganic Hg(II) was added to the mineral suspension in the anaerobic glovebox. All the other conditions were the same as those described above for MeHg degradation. Blank control experiments were performed in the same manner, containing borate buffer (20 mM) and 25 nM MeHg or inorganic Hg(II). At predetermined time intervals (0–3 h), sacrificial sample vials were taken out and analyzed for Hg and Fe species. All experiments were run in duplicate.

2.3. Analysis

Degradation of MeHg and its product inorganic Hg species distributions were determined. For the analysis of dissolved MeHg ($MeHg_{sol}$) and dissolved inorganic Hg(II) (Hg_{sol}) at selected time points, 1 mL of suspension was immediately mixed with 0.2 mM of methanol to stop the further oxidation of MeHg by $^{\cdot}OH$, and the concentrations of $MeHg_{sol}$ and Hg_{sol} were then determined after centrifugation and filtration. For the analysis of total MeHg ($MeHg_{total}$) and non-purgeable Hg(II) (Hg_{NP}), 2,3-dimercapto-1-propanesulfonic acid (DMPS), a strong mercury chelating agent (Liu et al., 2016), was added to sample vials to yield a final concentration of 0.1 mM and mixed for an additional 10 min before centrifugation and filtration. Then, adsorbed MeHg ($MeHg_{ads}$) and adsorbed inorganic Hg(II) (Hg_{ads}) on minerals could be calculated by the difference between the measured contents of MeHg and Hg(II) in filtrate solutions with and without DMPS washing protocol (that is, $MeHg_{ads} = MeHg_{total} - MeHg_{sol}$, and $Hg_{ads} = Hg_{NP} - Hg_{sol}$). Besides, total mercury (Hg_{Total}) was analyzed via blank control experiments with only MeHg and buffer solution. The recovery of blank controls was within 99.1 ± 4.1 % for a 25 nM control solution made in the assay medium, and nondetectable levels of inorganic mercury were produced, indicating that no MeHg degradation was induced without the presence of Fe (II)-bearing minerals. For the analysis of gaseous Hg(0), sacrificial vials were purged with purified air and purgeable Hg(0) were collected in trapping solutions (0.6 % $KMnO_4$ + 2.5 % HNO_3 + 2.5 % H_2SO_4). All filtered samples were determined for both MeHg and inorganic Hg(II) as soon as possible by online high-performance liquid chromatography-ultraviolet digestion-atomic fluorescence spectrometry hyphenated systems (HPLC-UV-AFS) (Xie et al., 2023). Mercury standard references and blanks were run every 15–20 samples, with detection limits of 0.036 μg/L for MeHg and 0.067 μg/L for inorganic Hg(II).

For the analysis of Fe species, adsorbed Fe on minerals was extracted by 0.5 M HCl. The concentration of dissolved Fe(II) was measured by the ferrozine method (Viollier et al., 2000). Mineral structural Fe(II) was measured by a modified HF-H_2SO_4/phenanthroline digestion method (Luan and Burgos, 2012). Total Fe (Fe_T) was determined by reducing Fe (III) through the addition of 10 % hydroxylamine hydrochloride.

To measure the cumulative concentration of $^{\cdot}OH$ generated from the oxidation of rNAu-2 (10 g/L), 10 mM sodium benzoate (BA) instead of mercury was added to the suspension. The hydroxylation of BA to *p*-hydroxybenzoic acid (*p*-HBA) by $^{\cdot}OH$ has been documented to estimate the production of $^{\cdot}OH$ (Cheng et al., 2020b). The concentration of *p*-HBA was then measured by an HPLC instrument equipped with a UV detector at a detection wavelength of 225 nm. Additionally, electron spin resonance (ESR) assay was conducted to confirm the production of $^{\cdot}OH$ and $O_2^{\cdot-}$. During the oxidation of Fe(II)-bearing minerals, a 100-μL sample was mixed with 100 μL of 0.1 M 5,5-dimethyl-1-pyrroline-N-oxide (DMPO) to form a DMPO-$^{\cdot}OH$ adduct, and a 50-μL sample was mixed with 50 μL of 0.1 M DMPO to form an DMPO—$O_2^{\cdot-}$ adduct. The mixtures were then measured on a Bruker EMX ESR spectrometer with a microwave bridge (receiver gain, 1000; modulation amplitude, 1 G; microwave power, 19.61 mW; modulation frequency, 9.85 GHz; center field:

3510 G).

3. Results and discussion

3.1. Degradation of MeHg induced by oxidation of Fe(II)-bearing minerals

To evaluate whether degradation of MeHg could be induced by oxidation of Fe(II)-bearing minerals, we prepared mineral structural Fe (II) and surface-adsorbed Fe(II), and monitored the variation of total MeHg when experimental assays were exposed to air. Near half of MeHg was degraded during the oxidation of reduced nontronite NAu-2 (rNAu-2, 2 g/L, 3.18 mM Fe(II)) (Fig. 1a). Additionally, when 2 mM Fe(II) was pre-adsorbed on Fe_3O_4 surfaces, ~30 % MeHg degradation was noticed during oxidation process (Fig. 1c). In contrast, under anoxic conditions, only a small percentage of MeHg exhibited a decrease (Fig. 1a and c). This limited decrease in total MeHg suggests that both mineral structural Fe(II) and surface-adsorbed Fe(II) have limited capacity to degrade MeHg directly in a reducing environment.

Along with drastic MeHg degradation under oxic conditions, oxidation of both structural Fe(II) in rNAu-2 and adsorbed Fe(II) on Fe_3O_4 surfaces were observed as well (Fig. 1b and d). Furthermore, no degradation of MeHg was assessed when NAu-2 had not been reduced (Fig. S2). Similarly, negligible decrease of MeHg was observed with Fe_3O_4 alone or with soluble Fe(II) alone (Fig. S3). Taken together with the fact that MeHg remained comparatively constant under anoxic conditions, these results collectively imply that MeHg degradation was likely induced by the oxidation of structural Fe(II) in rNAu-2 or adsorbed Fe(II) on Fe_3O_4 surfaces.

The relationship between MeHg degradation and Fe(II) oxidation was further confirmed during the oxidation of Fe(II)-bearing minerals. Oxidation of both mineral structural Fe(II) and surface-adsorbed Fe(II) exhibited strong correlations with degradation of MeHg under dark oxic conditions ($R^2 = 0.89$, Fig. 2a), suggesting that MeHg degradation was induced by Fe(II) oxidation. Notably, greater MeHg degradation and a higher reaction rate were noticed during the oxidation of rNAu-2 than Fe_3O_4-Fe(II) (Fig. 2b). However, the concentration of oxidized Fe(II) in rNAu-2 (~0.89 mM) was nearly equal to that in Fe_3O_4-Fe(II) (~0.90 mM) after 3-h oxidation (Fig. 2a). These results demonstrated that MeHg degradation efficiency was likely dependent on the type of Fe(II) species, and mineral structural Fe(II) in rNAu-2 ($k = 57.30$, Fig. 2a) was more reactive than adsorbed Fe(II) on Fe_3O_4 surfaces ($k = 28.01$, Fig. 2a). This observation aligns with previous findings that mineral structural Fe(II) plays a more significant role than surface-adsorbed Fe(II) in contaminant removal during sediment oxygenation (Zhang et al., 2023b). Collectively, these results illustrate that abiotic degradation of MeHg under dark oxic conditions was induced by the oxidation of Fe (II)-bearing minerals.

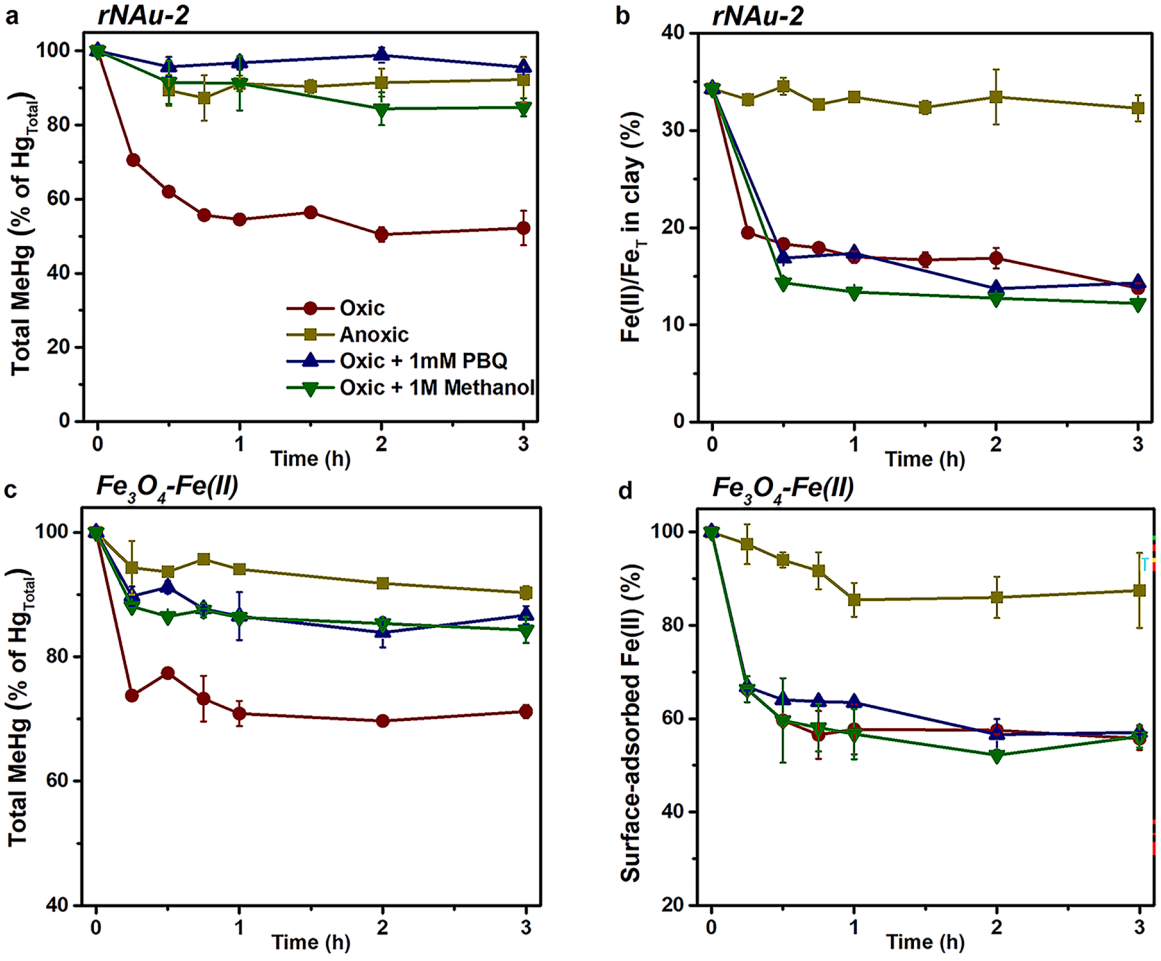


Fig. 1. Methylmercury degradation coupled to Fe(II) oxidation during oxidation of Fe(II)-bearing minerals. a-b, Analysis of total MeHg (a) and the ratio of Fe(II)/Fe_T in clay (b) over time during reactions between MeHg (25 nM) and rNAu-2 (2 g/L, 3.18 mM Fe(II)) in the presence of O_2. c-d, Analysis of total MeHg (c) and the ratio of adsorbed Fe(II) on Fe_3O_4 surfaces (d) over time during reactions between MeHg (25 nM) and surface-adsorbed Fe(II) (2 mM) in the presence of O_2. Surface-adsorbed Fe(II) was prepared by mixing 2 mM dissolved Fe(II) with 50 g/L Fe_3O_4 overnight. Similar experiments were carried out with the addition of hydroxyl radical scavenger methanol (1 M) or superoxide scavenger PBQ (1 mM). Anoxic experiments in the absence of O_2 were used as controls. Each point and error bar represents the mean and standard deviation of duplicate experiments.

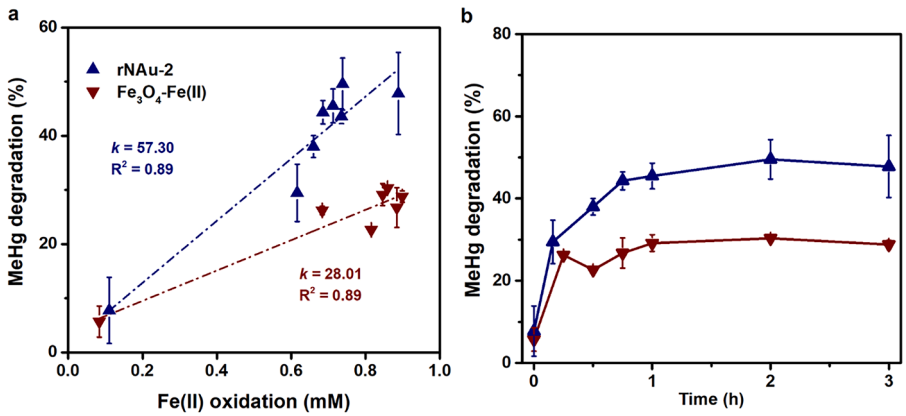


Fig. 2. Methylmercury degradation during oxidation of different Fe(II) species. a, Relationship between MeHg degradation and Fe(II) oxidation upon oxidation of mineral structural Fe(II) (2 g/L rNAu-2) and surface-adsorbed Fe(II) (2 mM) under dark oxic conditions. b, The amounts of MeHg degradation induced by oxidation of rNAu-2 and Fe_3O_4-Fe(II). Each point and error bar represents the mean and standard deviation of duplicate experiments.

3.2. Mechanism of MeHg degradation under dark oxic conditions

Since hydroxyl radical (˙OH), known for its extremely high oxidation ability, has been recognized as the main reactive species produced from the oxidation of Fe(II)-bearing minerals (Cheng et al., 2020b; Yuan et al., 2018; Zeng et al., 2020), it is likely that the degradation of MeHg is also induced by ˙OH generated from the oxidation of Fe(II)-bearing mineral under dark oxic conditions. Therefore, experiments were conducted to explore the involvement of ˙OH in the mechanism of MeHg degradation under dark oxic conditions. During the oxidation of rNAu-2 and Fe_3O_4-Fe(II), the production of ˙OH was evident, as indicated by the observation of characteristic 1:2:2:1 peaks in ESR spectrum (Fig. 3a). Additionally, the cumulative concentration of ·OH rapidly increased to 99.8 μM following a 3-hour oxidation of rNAu-2, as demonstrated by employing benzoate as a trapping agent (Fig. S4). However, the peaks in ESR spectrum disappeared in the presence of methanol, an effective scavenger for ˙OH (Fig. 3a), and negligible ˙OH production was noticed with the addition of methanol (Fig. S4). Considering the confirmed quenching effect of methanol on the generation of ·OH, a significant inhibition of MeHg degradation was expectedly noticed in the presence of methanol during the oxidation of Fe(II)-bearing minerals (Fig. 1a and c). The addition of methanol resulted in a decrease of approximately 70 % and 50 % in MeHg degradation after the oxidation of rNAu-2 and Fe_3O_4-Fe(II), respectively. Furthermore, comparable amounts of Fe(II) were oxidized regardless of the presence of methanol (Fig. 1b and d), suggesting that the influence of ˙OH scavenger on the overall oxidation of Fe(II) was slight. Consequently, the suppression of MeHg degradation caused by ˙OH quenching confirmed the contribution of ˙OH generated from the oxidation of Fe(II)-bearing minerals (Fig. 4).

The quenching ˙OH did not completely inhibit MeHg degradation (Fig. 1a,c), indicating the involvement of other reactive species in this process. Previous studies have reported that superoxide radical ($O_2^{\cdot-}$) is

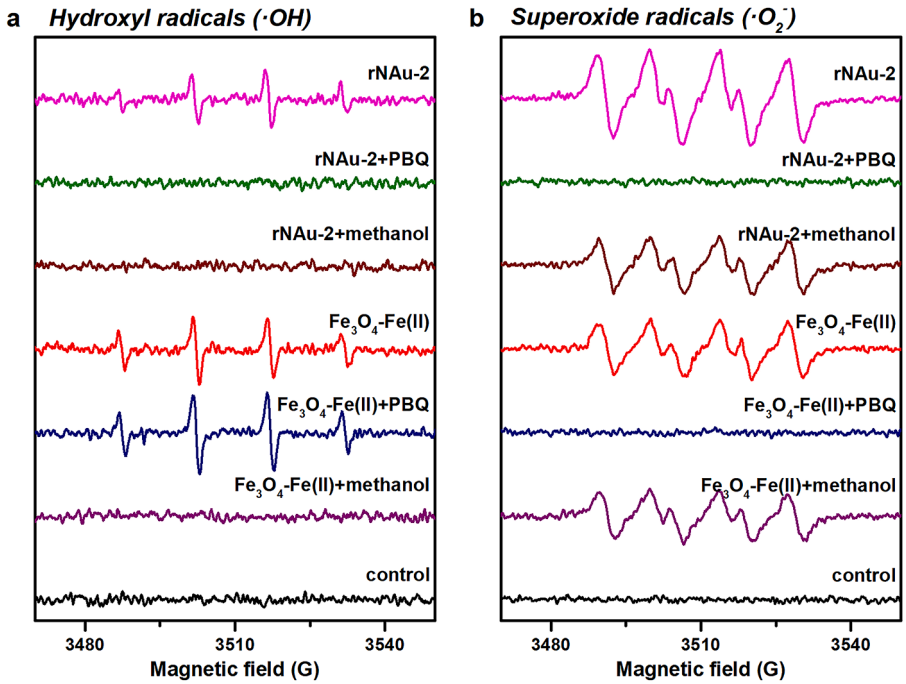


Fig. 3. Qualitative assessment of hydroxyl radicals (˙OH) and superoxide radicals ($O_2^{\cdot-}$) during oxidation of Fe(II)-bearing minerals. Electron spin resonance (ESR) spectrum of ˙OH (a) and $O_2^{\cdot-}$ (b) during oxidation of both mineral structural Fe(II) (2 g/L rNAu-2) and surface-adsorbed Fe(II) (2 mM). For ESR assays, rNAu-2 and Fe_3O_4-Fe(II) were oxygenated for 3 min under dark conditions before adding DMPO solutions. Similar experiments were carried out with the addition of hydroxyl radical scavenger methanol (1 M) or superoxide scavenger PBQ (1 mM). Experiments with only MeHg and buffer solution were used as controls.

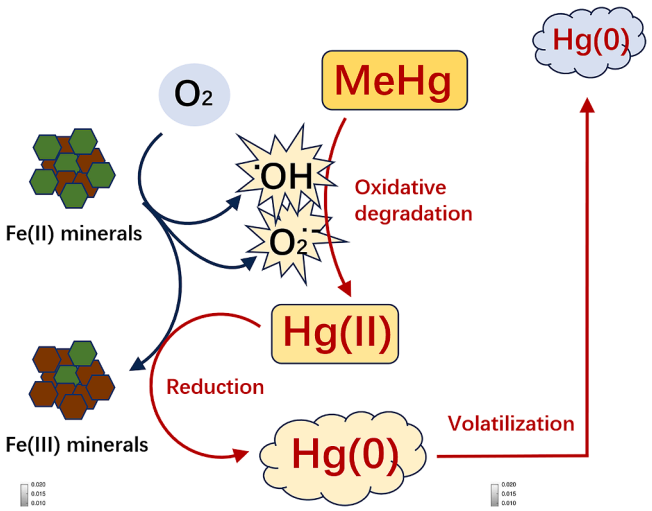


Fig. 4. Conceptual illustration of MeHg demethylation coupled with Hg(0) generation induced by the oxidation of Fe(II)-bearing minerals. MeHg is oxidatively degraded into Hg(II) by generated $^{\cdot}OH$ and $O_2^{\cdot-}$ during the oxidation of Fe(II)-bearing minerals. The produced Hg(II) is promptly reduced by Fe(II)-bearing minerals, leading to the generation and subsequent volatilization of Hg(0).

capable of converting MeHg to inorganic Hg (Mailloux et al., 2015; Yasutake and Hirayama, 2001; Zhang et al., 2023a). In this study, the generation of $O_2^{\cdot-}$ was confirmed by ESR during the oxidation of Fe (II)-bearing minerals (Fig. 3b). Upon addition of 1,4-benzoquinone (PBQ), a known scavenger of $O_2^{\cdot-}$ (Han et al., 2017), no measurable signals were observed (Fig. 3b), and the addition of PBQ eliminated MeHg degradation during the oxidation of Fe(II)-bearing minerals as well (Fig. 1a and c). Little effect on Fe(II) oxidation was observed in the presence of PBQ (Fig. 1b and d). During the oxidation of rNAu-2, a clear inhibition of $^{\cdot}OH$ production was observed when $O_2^{\cdot-}$ was quenched by the addition of PBQ (Fig. 3a and Fig. S4), indicating that $O_2^{\cdot-}$ served as the intermediate for $^{\cdot}OH$ production via one-electron transfer mechanism (Han et al., 2022; Yuan et al., 2018). Interestingly, the addition of $^{\cdot}OH$ scavenger methanol inhibited around 68 % of MeHg degradation, whereas the addition of $O_2^{\cdot-}$ scavenger PBQ suppressed over 90 % of MeHg degradation (Fig. 1a). These results demonstrate that in addition to serving as an intermediate for $^{\cdot}OH$ production to indirectly participate in MeHg degradation, $O_2^{\cdot-}$ may directly degrade MeHg during the oxidation of mineral structural Fe(II). Furthermore, during the oxidation of Fe_3O_4-Fe(II), despite the negligible effect on $^{\cdot}OH$ production when $O_2^{\cdot-}$ was quenched (Fig. 3a), there was still approximately a 56 % decrease in MeHg degradation (Fig. 1c). These results suggest that $^{\cdot}OH$ is primarily generated via the two-electron transfer mechanism without requiring $O_2^{\cdot-}$ as an intermediate during the oxidation of surface-adsorbed Fe(II) (Cheng et al., 2016). Importantly, $O_2^{\cdot-}$ is capable of direct degradation of MeHg. Overall, both $^{\cdot}OH$ and $O_2^{\cdot-}$ are responsible for the abiotic degradation of MeHg during the oxidation of Fe(II)-bearing minerals under dark oxic conditions (Fig. 4).

3.3. The simultaneous oxidative degradation of MeHg and reduction of Hg(II) during the oxidation of Fe(II)-bearing minerals

Reactive oxygen species generated from the oxidation of Fe(II)-bearing minerals are generally recognized as the primary agents responsible for the oxidative degradation of both inorganic and organic contaminants (Chen et al., 2019; Cheng et al., 2016). Thus, MeHg degradation induced by the oxidation of Fe(II)-bearing minerals in this study was supposed to occur oxidatively, leading to the production of inorganic Hg(II). However, upon the oxidation of rNAu-2, only 6.7 % Hg (II) was initially observed in solution within the first 15 min, and it diminished immediately to undetectable levels (Fig. 5a). Furthermore, no Hg(II) was found to be adsorbed on rNAu-2 throughout the experiment (Fig. 5a). Similarly, minimal Hg(II) was detected during the oxidation of surface-adsorbed Fe(II) (Fig. 5c). These results strongly demonstrate that inorganic Hg(II) was not the primary product, despite the oxidative degradation of MeHg induced by the oxidation of both rNAu-2 and Fe_3O_4-Fe(II).

To further explore the mechanism of abiotic MeHg demethylation during the oxidation of Fe(II)-bearing minerals, we quantified gaseous Hg(0), another form of inorganic mercury. Purging and trapping experiments revealed that approximately 48.2 % and 35.4 % Hg(0) were generated during the oxidation of rNAu-2 and Fe_3O_4-Fe(II), respectively (Fig. S5). Correspondingly, around 47.8 % and 30.3 % MeHg were degraded in the cases of rNAu-2 and Fe_3O_4-Fe(II), respectively (Fig. 1a and c). The comparable amounts of Hg(0) production and MeHg degradation demonstrated that nearly all inorganic Hg species resulting from MeHg demethylation during the oxidation of Fe(II)-bearing minerals were in the form of Hg(0) rather than Hg(II) (Fig. 4). Given that the losses of Hg_{Total} in batch experiments were primarily attributed to the production of purgeable Hg(0), the production of Hg(0) at different time points during oxidation process was determined by subtracting MeHg and Hg(II) from Hg_{Total}. Interestingly, Hg(0) production during the oxidation of both rNAu-2 and Fe_3O_4-Fe(II) was observed to initially increase quickly and then remain almost constant (Fig. 5a and c).

Considering the transient Hg(II) production during the oxidation of rNAu-2 (Fig. 5a), the oxidative degradation of MeHg, followed by the prompt reduction of the produced Hg(II), may lead to Hg(0) generation. Additionally, given that ferrous iron in both rNAu-2 and Fe_3O_4-Fe(II) was not completely oxidized after 3-h oxidation (Fig. 1b and d), we hypothesized that Fe(II)-bearing minerals, known for their capability to reduce Hg(II) (Bone et al., 2014; O'Loughlin et al., 2020), were responsible for the reduction of Hg(II) and subsequent Hg(0) production. Batch experiments were carried out to confirm the ability of both rNAu-2 and Fe_3O_4-Fe(II) to reduce Hg(II). Approximately 91.1 % and 30.7 % of the Hg(II) were rapidly reduced by rNAu-2 and Fe_3O_4-Fe(II) (Fig. 5b and d). Furthermore, purging and trapping experiments revealed that approximately 90.2 % and 30.0 % of the Hg(0) produced in the Hg(II) reduction experiments were detected (Fig. S5), aligning with the previously mentioned Hg_{Total} losses in the Hg(II) reduction experiments. Taken together, during the oxidation of Fe(II)-bearing minerals, MeHg was initially oxidatively degraded to inorganic Hg(II), and the generated Hg(II) was promptly reduced to Hg(0) by Fe(II)-bearing minerals (Fig. 4).

3.4. Environmental implications

This newly discovered dark abiotic process of MeHg demethylation in this study is significant due to the widespread presence of Fe(II)-bearing minerals and redox-dynamic environments. These environments are subject to O_2 perturbation in the subsurface, which can result from both natural processes (i.e., groundwater table fluctuations in aquifers (Zhang et al., 2020), wetlands, and coasts (Hsieh et al., 2015)) and artificial processes (i.e., air-based remediation (Rabideau et al., 1999; Yang et al., 2005) and farming activities (Wang et al., 2023)). Paddy soils, emerging as a new hotspot for MeHg contamination (Zhang et al., 2010a, 2010b), are recognized as typical iron rich, redox-dynamic environments with cyclic flooding and draining. The demethylation of MeHg and subsequent generation of Hg(0) under such fluctuations reduce the bioavailability of MeHg to rice crops, thereby relieving the health risks to humans via natural attenuation. Notably, this novel demethylation mechanism of MeHg holds practical implications for engineering applications. For on-site remediation of MeHg-contaminated groundwater, air-sparging treatment in iron (II)-rich environments could induce redox fluctuations to trigger the light-independent detoxification of MeHg. Additionally, in wetlands affected by periodic tides, supplementation of iron(II) minerals or

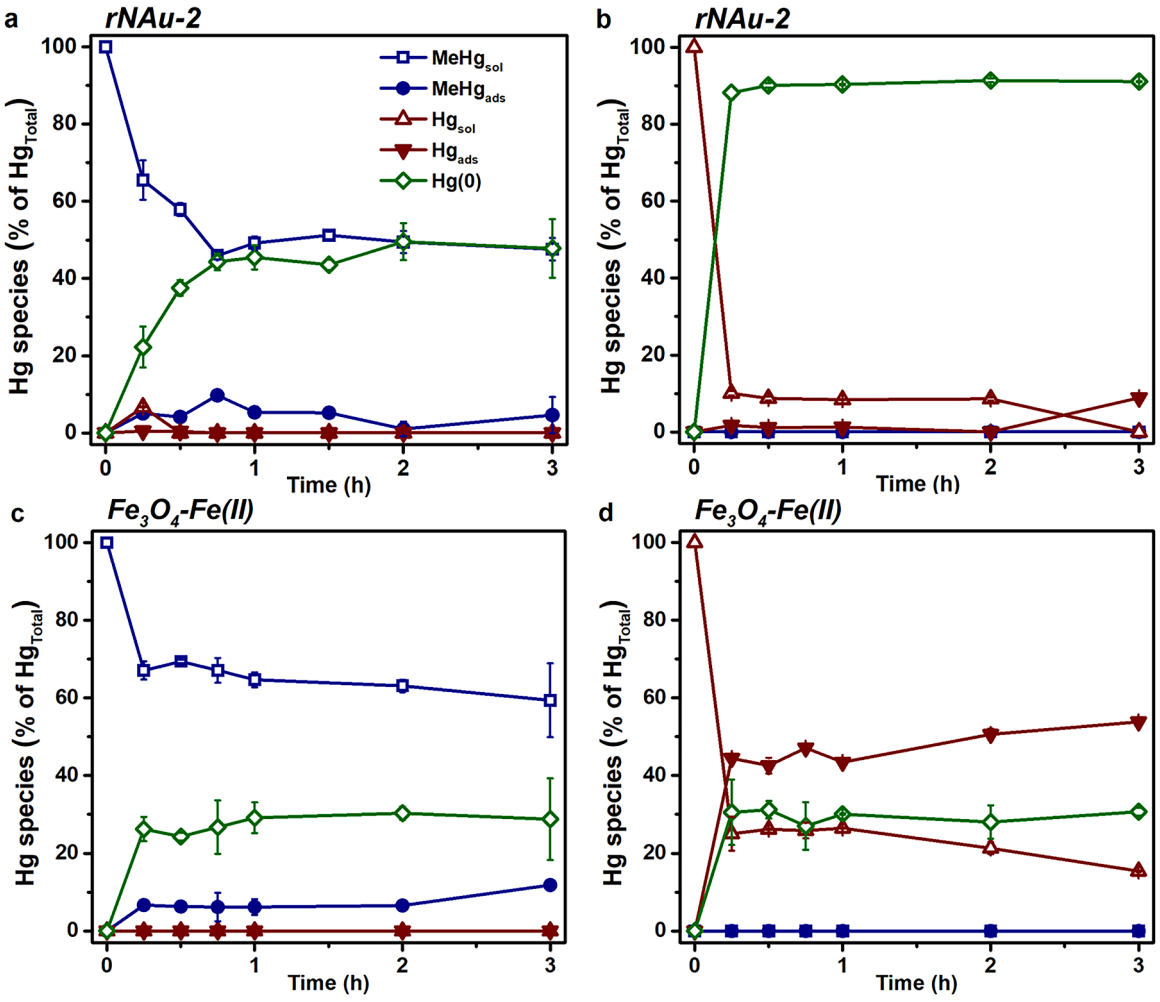


Fig. 5. Analysis of mercury (Hg) species during MeHg degradation and Hg(II) reduction by Fe(II)-bearing minerals. a,b, Analysis of Hg species during MeHg degradation (a) and Hg(II) reduction (b) induced by 2 g/L rNAu-2. c,d, Analysis of Hg species during MeHg degradation (c) and Hg(II) reduction (d) induced by 2 mM surface-adsorbed Fe(II). The degradation experiments of MeHg were started by moving the systems out from the glovebox to air, and the reduction experiments of Hg (II) were conducted in glovebox under dark conditions. Soluble MeHg, adsorbed MeHg, soluble inorganic Hg(II), and adsorbed inorganic Hg(II), were referred as $MeHg_{sol}$, $MeHg_{ads}$, Hg_{sol}, and Hg_{ads}, respectively. Each point and error bar represents the mean and standard deviation of duplicate experiments.

iron-reducing bacteria may mitigate risks by facilitating the natural attenuation of MeHg.

4. Conclusion

In this study, we have discovered that demethylation of MeHg occurs in the presence of Fe(II)-bearing minerals under dark oxic conditions. This phenomenon may be largely underestimated or even overlooked owing to the prevailing belief that abiotic degradation of MeHg mainly occurs in surface sunlit waters. $^{\cdot}OH$ and $O_2^{\cdot-}$, produced from the oxidation of Fe(II)-bearing minerals, are responsible for oxidative degradation of MeHg and subsequent production of inorganic Hg(II). Surprisingly, we also observed immediate reduction of Hg(II) by Fe(II)-bearing minerals occurs under such oxic conditions, resulting in the generation of gaseous Hg(0). The oxidative degradation of MeHg coupled with Hg(II) reduction highlights the significant role of Fe(II)-bearing minerals in potential detoxification pathways for MeHg, especially in O_2-perturbed environments. Our research findings contribute to understanding the light-independent abiotic cycling of mercury in redox-dynamic environments and offer novel perspectives on the natural attenuation and remediation of MeHg in subsurface environmental compartments.

CRediT authorship contribution statement

Fuyu Xie: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Qingke Yuan:** Writing – review & editing, Validation. **Ying Meng:** Writing – review & editing, Validation, Conceptualization. **Fubo Luan:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was supported by the National Natural Science Foundation of China (No. 41977280 and U1904205).

References

Amirbahman, A., Kent, D.B., Curtis, G.P., Marvin-DiPasquale, M.C., 2013. Kinetics of homogeneous and surface-catalyzed mercury(II) reduction by iron(II). Environ. Sci. Technol. 47 (13), 7204–7213.

Barkay, T., Gu, B., 2022. Demethylation-the other side of the mercury methylation coin: a critical review. ACS Environ. Au 2 (2), 77–97.

Blum, J.D., 2011. Marine mercury breakdown. Nat. Geosci. 4 (3), 139–140.

Bone, S.E., Bargar, J.R., Sposito, G., 2014. Mackinawite (FeS) reduces mercury(II) under sulfidic conditions. Environ. Sci. Technol. 48 (18), 10681–10689.

Chen, C., Kukkadapu, R.K., Lazareva, O., Sparks, D.L., 2017. Solid-phase Fe speciation along the vertical redox gradients in floodplains using XAS and Mössbauer spectroscopies. Environ. Sci. Technol. 51 (14), 7903–7912.

Chen, N., Fang, G., Liu, G., Zhou, D., Gao, J., Gu, C., 2018. The effects of Fe-bearing smectite clays on OH formation and diethyl phthalate degradation with polyphenols and H2O2. J. Hazard. Mater. 357, 483–490.

Chen, N., Fang, G., Liu, G., Zhou, D., Gao, J., Gu, C., 2019. The degradation of diethyl phthalate by reduced smectite clays and dissolved oxygen. Chem. Eng. J. 355, 247–254.

Cheng, D., Liao, W., Yuan, S., 2020a. Effect of in situ generated iron oxyhydroxide coatings on FeS oxygenation and resultant hydroxyl radical production for contaminant degradation. Chem. Eng. J. 394, 124961.

Cheng, D., Neumann, A., Yuan, S., Liao, W., Qian, A., 2020b. Oxidative degradation of organic contaminants by FeS in the presence of O2. Environ. Sci. Technol. 54 (7), 4091–4101.

Cheng, D., Yuan, S., Liao, P., Zhang, P., 2016. Oxidizing impact induced by mackinawite (FeS) nanoparticles at oxic conditions due to production of hydroxyl radicals. Environ. Sci. Technol. 50 (21), 11646–11653.

Driscoll, C.T., Mason, R.P., Chan, H.M., Jacob, D.J., Pirrone, N., 2013. Mercury as a global pollutant: sources, pathways, and effects. Environ. Sci. Technol. 47 (10), 4967–4983.

Han, R., Wang, Z., Lv, J., Zhu, Z., Yu, G.H., Li, G., Zhu, Y.G., 2022. Multiple effects of humic components on microbially mediated iron redox processes and production of hydroxyl radicals. Environ. Sci. Technol. 56 (22), 16419–16427.

Han, X., Li, Y., Li, D., Liu, C., 2017. Role of free radicals/reactive oxygen species in MeHg photodegradation: importance of utilizing appropriate scavengers. Environ. Sci. Technol. 51 (7), 3784–3793.

Hsieh, P.C., Hsu, H.T., Liao, C.B., Chiueh, P.T., 2015. Groundwater response to tidal fluctuation and rainfall in a coastal aquifer. J. Hydrol. 521, 132–140.

Hu, H., Lin, H., Zheng, W., Tomanicek, S.J., Johs, A., Feng, X., Elias, D.A., Liang, L., Gu, B., 2013. Oxidation and methylation of dissolved elemental mercury by anaerobic bacteria. Nat. Geosci. 6 (9), 751–754.

Jeong, H.Y., Sun, K., Hayes, K.F., 2010. Microscopic and spectroscopic characterization of Hg(II) immobilization by mackinawite (FeS). Environ. Sci. Technol. 44 (19), 7476–7483.

Kang-Yun, C.S., Liang, X., Dershwitz, P., Gu, W., Schepers, A., Flatley, A., Lichtmannegger, J., Zischka, H., Zhang, L., Lu, X., Gu, B., Ledesma, J.C., Pelger, D.J., DiSpirito, A.A., Semrau, J.D., 2022. Evidence for methanobactin "Theft" and novel chalkophore production in methanotrophs: impact on methanotrophic-mediated methylmercury degradation. ISME J. 16 (1), 211–220.

Kappler, A., Bryce, C., Mansor, M., Lueder, U., Byrne, J.M., Swanner, E.D., 2021. An evolving view on biogeochemical cycling of iron. Nat. Rev. Microbiol. 19 (6), 360–374.

Kritee, K., Barkay, T., Blum, J.D., 2009. Mass dependent stable isotope fractionation of mercury during *mer* mediated microbial degradation of monomethylmercury. Geochim. Cosmochim. Acta 73 (5), 1285–1296.

Liu, J., Chen, J., Poulain, A.J., Pu, Q., Hao, Z., Meng, B., Feng, X., 2023. Mercury and sulfur redox cycling affect methylmercury levels in rice paddy soils across a contamination gradient. Environ. Sci. Technol. 57 (21), 8149–8160.

Liu, M., Zhang, Q., Cheng, M., He, Y., Chen, L., Zhang, H., Cao, H., Shen, H., Zhang, W., Tao, S., Wang, X., 2019. Rice life cycle-based global mercury biotransport and human methylmercury exposure. Nat. Commun. 10 (1), 5164.

Liu, Y., Lu, X., Zhao, L., An, J., He, J., Pierce, E.M., Johs, A., Gu, B., 2016. Effects of cellular sorption on mercury bioavailability and methylmercury production by *Desulfovibrio desulfuricans* ND132. Environ. Sci. Technol. 50 (24), 13335–13341.

Lu, X., Gu, W., Zhao, L., Ul Haque, M.F., DiSpirito, A.A., Semrau, J.D., Gu, B., 2017. Methylmercury uptake and degradation by methanotrophs. Sci. Adv. 3 (5), 1–5.

Luan, F., Burgos, W.D., 2012. Sequential extraction method for determination of Fe and U in suspensions of iron-bearing phyllosilicates and uranium. Environ. Sci. Technol. 46, 11995–12002.

Luan, F., Liu, Y., Griffin, A.M., Gorski, C.A., Burgos, W.D., 2015. Iron(III)-bearing clay minerals enhance bioreduction of nitrobenzene by *Shewanella putrefaciens* CN32. Environ. Sci. Technol. 49, 1418–1426.

Mailloux, R.J., Yumvihoze, E., Chan, H.M., 2015. Superoxide anion radical (O2-) degrades methylmercury to inorganic mercury in human astrocytoma cell line (CCF-STTG1). Chem. Biol. Interact. 239, 46–55.

O'Loughlin, E.J., Boyanov, M.I., Kemner, K.M., Thalhammer, K.O., 2020. Reduction of Hg(II) by Fe(II)-bearing smectite clay minerals. Minerals 10 (12), 1–13.

Parks, J.M., Johs, A., Podar, M., Bridou, R., Hurt, R.A.J., Smith, S.D., Tomanicek, S.J., Qian, Y., Brown, S.D., Brandt, C.C., Palumbo, A.V., Smith, J.C., Wall, J.D., Elias, D. A., Liang, L., 2013. The genetic basis for bacterial mercury methylation. Science 339 (6125), 1332–1335.

Peretyazhko, T.S., Zachara, J.M., Kukkadapu, R.K., Heald, S.M., Kutnyakov, I.V., Resch, C.T., Arey, B.W., Wang, C.M., Kovarik, L., Phillips, J.L., Moore, D.A., 2012. Pertechnetate (TcO4−) reduction by reactive ferrous iron forms in naturally anoxic, redox transition zone sediments from the Hanford Site, USA. Geochim. Cosmochim. Acta 92, 48–66.

Qian, Y., Yin, X., Lin, H., Rao, B., Brooks, S.C., Liang, L., Gu, B., 2014. Why dissolved organic matter enhances photodegradation of methylmercury. Environ. Sci. Technol. Lett. 1 (10), 426–431.

Rabideau, A.J., Blayden, J.M., Ganguly, C., 1999. Field performance of air-sparging system for removing TCE from groundwater. Environ. Sci. Technol. 33 (1), 157–162.

Sellers, P., Kelly, C.A., Ruddt, J.W.M., MacHutchont, A.R., 1996. Photodegradation of methylmercury in lakes. Nature 380 (25), 694–697.

Sun, Z., Huang, M., Liu, C., Fang, G., Chen, N., Zhou, D., Gao, J., 2020. The formation of ·OH with Fe-bearing smectite clays and low-molecular-weight thiols: implication of As(III) removal. Water Res. 174, 115631.

Tai, C., Li, Y., Yin, Y., Scinto, L.J., Jiang, G., Cai, Y., 2014. Methylmercury photodegradation in surface water of the Florida Everglades: importance of dissolved organic matter-methylmercury complexation. Environ. Sci. Technol. 48 (13), 7333–7340.

Tian, L., Guan, W., Ji, Y., He, X., Chen, W., Alvarez, P.J.J., Zhang, T., 2021. Microbial methylation potential of mercury sulfide particles dictated by surface structure. Nat. Geosci. 14, 409–416.

Trasande, L., Landrigan, P.J., Schechter, C., 2005. Public health and economic consequences of methyl mercury toxicity to the developing brain. Environ. Health Perspect. 113 (5), 590–596.

Viollier, E., Inglett, P.W., Hunter, K., Roychoudhury, A.N., Cappellen, P.V., 2000. The ferrozine method revisited: Fe(II)/Fe(III) determination in natural waters. Appl. Geochem. 15, 785–790.

Wang, Y., Huang, D., Ge, C., Wang, X., Zhu, C., Chen, N., Fang, G., Zhou, D., 2023. Amendment of organic acids significantly enhanced hydroxyl radical production during oxygenation of paddy soils. J. Hazard. Mater. 457, 131799.

Wiatrowski, H.A., Das, S., Kukkadapu, R., Ilton, E.S., Barkay, T., Yee, N., 2009. Reduction of Hg(II) to Hg(0) by magnetite. Environ. Sci. Technol. 43 (14), 5307–5313.

Xie, F., Yuan, Q., Qie, Y., Meng, Y., Luan, F., 2023. Capacity, stability and energy requirement of divalent mercury uptake by non-methylating/non-demethylating bacteria. J. Hazard. Mater. 450, 131074.

Xie, W., Yuan, S., Tong, M., Ma, S., Liao, W., Zhang, N., Chen, C., 2020. Contaminant degradation by ·OH during sediment oxygenation: dependence on Fe(II) species. Environ. Sci. Technol. 54 (5), 2975–2984.

Yang, X., Beckmann, D., Fiorenza, S., Niedermeier, C., 2005. Field study of pulsed air sparging for remediation of petroleum hydrocarbon contaminated soil and groundwater. Environ. Sci. Technol. 39 (18), 7279–7286.

Yasutake, A., Hirayama, K., 2001. Evaluation of methylmercury biotransformation using rat liver slices. Arch. Toxicol. 75 (7), 400–406.

Yuan, S., Liu, X., Liao, W., Zhang, P., Wang, X., Tong, M., 2018. Mechanisms of electron transfer from structrual Fe(II) in reduced nontronite to oxygen for production of hydroxyl radicals. Geochim. Cosmochim. Acta 223, 422–436.

Zeng, Q., Dong, H., Wang, X., Yu, T., Cui, W., 2017. Degradation of 1, 4-dioxane by hydroxyl radicals produced from clay minerals. J. Hazard. Mater. 331, 88–98.

Zeng, Q., Wang, X., Liu, X., Huang, L., Hu, J., Chu, R., Tolic, N., Dong, H., 2020. Mutual interactions between reduced Fe-bearing clay minerals and humic acids under dark, oxygenated conditions: hydroxyl radical generation and humic acid transformation. Environ. Sci. Technol. 54 (23), 15013–15023.

Zhang, D., Xiang, Y., Liu, G., Liang, L., Chen, L., Shi, J., Yin, Y., Cai, Y., Jiang, G., 2023a. Mechanism and controlling factors on rapid methylmercury degradation by ligand-enhanced Fenton-like reaction at circumneutral pH. Chemosphere 324, 138291.

Zhang, H., Feng, X., Larssen, T., Qiu, G., Vogt, R.D., 2010a. In inland China, rice, rather than fish, is the major pathway for methylmercury exposure. Environ. Health Perspect. 118 (9), 1183–1188.

Zhang, H., Feng, X., Larssen, T., Shang, L., Li, P., 2010b. Bioaccumulation of methylmercury versus inorganic mercury in rice grain. Environ. Sci. Technol. 44 (12), 4499–4504.

Zhang, N., Bu, X., Li, Y., Zhang, Y., Yuan, S., Wen, Z., Tong, M., Lin, L., 2020. Water table fluctuations regulate hydrogen peroxide production and distribution in unconfined aquifers. Environ. Sci. Technol. 54 (8), 4942–4951.

Zhang, P., Liu, J., Yu, H., Cheng, D., Liu, H., Yuan, S., 2023b. Kinetic models for hydroxyl radical production and contaminant removal during soil/sediment oxygenation. Water Res. 240, 120071.

Zhang, T., Hsu-Kim, H., 2010. Photolytic degradation of methylmercury enhanced by binding to natural organic ligands. Nat. Geosci. 3 (7), 473–476.

Zhao, Z., Meng, Y., Yuan, Q., Wang, Y., Lin, L., Liu, W., Luan, F., 2021. Microbial mobilization of arsenic from iron-bearing clay mineral through iron, arsenate, and simultaneous iron-arsenate reduction pathways. Sci. Total Environ. 763, 144613.